



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



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Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

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Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

- large enough to ignore discreteness noise, statistically sufficient number of particles

$$n\ell^3 \gg 1$$

- in collisional fluids, large enough to communicate local conditions, requires interactions

$$\ell \gg \lambda$$

Describing Fluid Equations

- *Eulerian View*: cells of small volumes in fixed grid, flow is defined for all $\mathbf{t}$ at position $\mathbf{r}$ with rate of change $\partial/\partial\mathbf{t}$ for any measurable quantity Q
- *Lagrangian View*: follow particular fluid elements, comoving reference system with flow evaluated in each one via $D/D\mathbf{t}$ as the rate of change
- *Translating Relation*: add convective derivative term to account for movement of fluid element

$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q$$

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)
- pressure p (scalar)
- temperature T (scalar)

■ *Conserved Quantities:*

- mass M (scalar)
- energy E (scalar)
- momentum $\mathbf{P}$ (vector)

■ *Derived Values:*

- viscosity η (scalar)
- vorticity $\boldsymbol{\omega}$ (vector)

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{u L}{\nu}$$

- $R < 1000$: Laminar Flow (reversible and ordered stream)
 - ▶ $R < 1$: Stokes Regime (without any separation)
 - ▶ $10 \lesssim R \lesssim 40$: Separation Zone (stable and stationary vortices)
- $R > 1000$: Turbulent Flow (stochastic and chaotic stream)
 - ▶ $40 \lesssim R \lesssim 4000$: Transitional Kármán Street (vortex shedding)
 - ▶ $R > 4000$: Kolmogorov Energy Cascade (fully developed phase)

Dimensionless Numbers

- *Mach Number*: ratio of bulk velocity magnitude to local sound speed

$$M = \frac{u}{c}$$

- $M \lesssim 0.3$: Incompressible (negligible density variations, pressure signals faster than flow)
- $M \approx 1$: Transonic (local excess of sound speed, weak shocks create drag)
- $M > 1$: Supersonic (thin front of discontinuity, bow shock heating)
- $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

Bulk Transport

- *Continuity Equation*: density ρ must increase where a fluid is compressed

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- *Linear Advection*: velocity field $\mathbf{v}$ is free of divergence for incompressible fluids

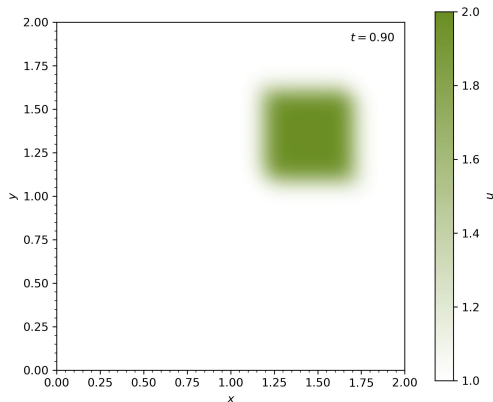
$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{u} = 0$$

- *Nonlinear Advection*: velocity field $\mathbf{u}$ is carried by itself instead of fixed background flow

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = 0$$

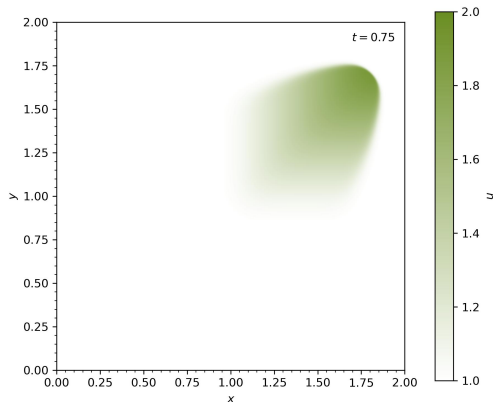
Linear Advection

- *Initialization:*
 - square wave
- *Boundaries:*
 - *Lower:* constant value
 - *Upper:* unconstrained
- *Observation:*
 - fixed profile
 - linear translation



Nonlinear Advection

- *Initialization:*
 - square wave
- *Boundaries:*
 - *Lower:* constant value
 - *Upper:* unconstrained
- *Observation:*
 - shock formation
 - rarefaction tail



Stochastic Transport

- *Diffusion*: kinematic viscosity ν describes spread due to random motion

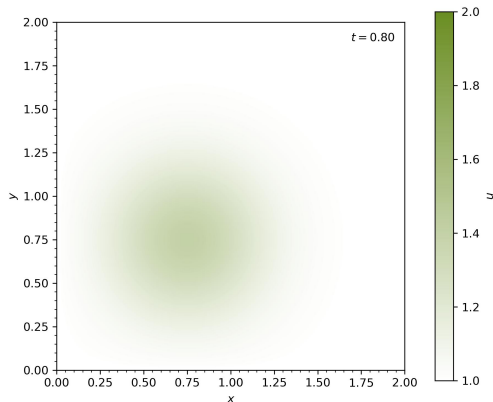
$$\frac{\partial u}{\partial t} = \nu \nabla^2 u$$

- *Bateman-Burgers Equation*: combines shock generation with counteracting diffusive term

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = \nu \nabla^2 u$$

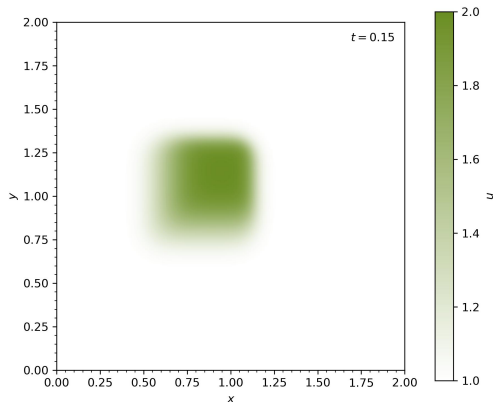
Diffusion

- *Initialization:*
 - square wave
- *Boundaries:*
 - *Both:* constant value
- *Observation:*
 - stationary decay
 - edge smoothing
 - spreading out



Bateman-Burgers Equation

- *Initialization:*
 - sawtooth wave
- *Boundaries:*
 - Both: periodic
- *Observation:*
 - nonlinear steepening
 - viscous smoothing
 - dissipating translation



Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

- Gravity: $\nabla^2 \Phi = 4\pi G \rho$
- Electrostatics: $\nabla^2 V = -\rho / \epsilon$
- Pressure: $\nabla^2 p = \rho \nabla \cdot ((u \cdot \nabla) u)$

Laplace Equation

- *Initialization:*

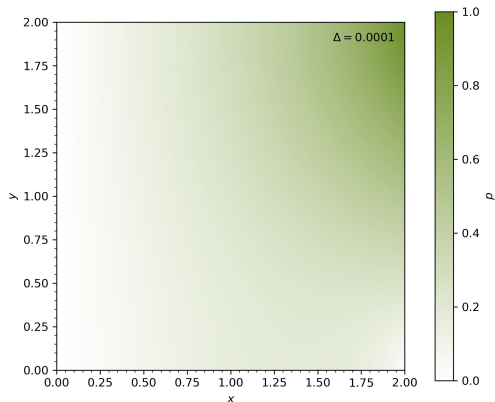
- empty field

- *Boundaries:*

- *Left:* fixed zero
- *Right:* linear slope
- *Top & Bottom:* no gradient

- *Observation:*

- iterative diffusion
- smooth relaxation



Poisson Equation

■ Initialization:

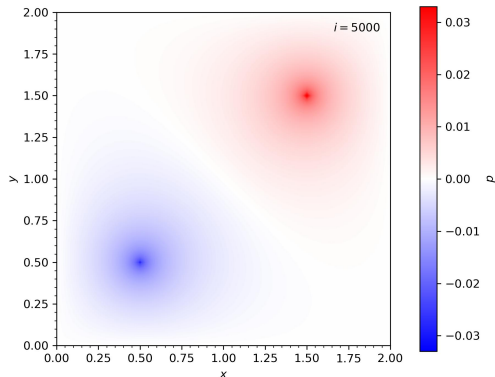
- empty field
- positive source
- negative sink

■ Boundaries:

- All: fixed zero

■ Observation:

- dipole field potential
- converging relaxation



Navier-Stokes Equations

- *Momentum Equation*: ensure $\nabla \cdot \mathbf{u} = 0$ by including ∇p to instantaneously cancel out any divergence

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}$$

- *Pressure Poisson*: require p to have relaxed to its equilibrium for all t before projecting

$$\nabla^2 p = \rho \nabla \cdot ((\mathbf{u} \cdot \nabla) \mathbf{u})$$

- fulfill at each step to accurately evolve initial configuration under given boundary conditions

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Lid Driven Cavity Flow

■ Initialization:

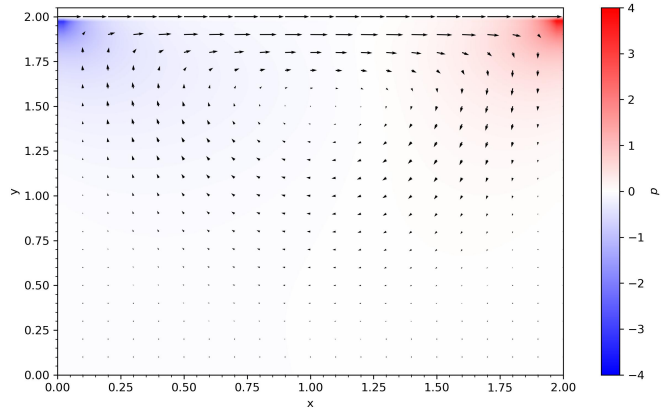
- vanishing velocity
- zero pressure

■ Boundaries:

- Top: $u = 1$ & $\nabla p = 0$
- Bottom: $u = 0$ & $p = 0$
- Left & Right: $u = 0$ & $\nabla p = 0$

■ Observation:

- vortex formation
- pressure gradient



Pressure Driven Channel Flow

■ Initialization:

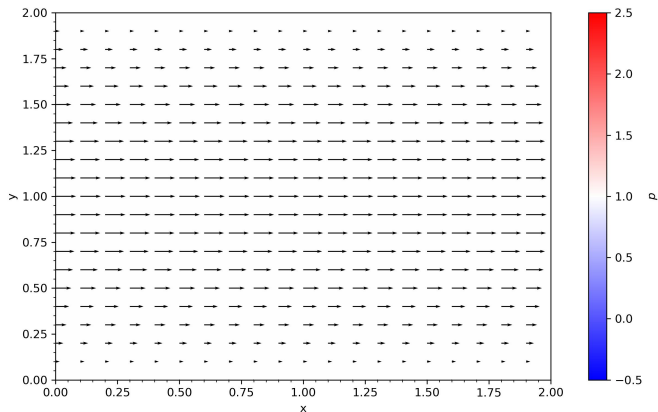
- vanishing velocity
- unity pressure
- constant source

■ Boundaries:

- Top & Bottom: $u = 0$ & $\nabla p = 0$
- Left & Right: periodic

■ Observation:

- curved velocity profile
- boundary layer formation



Locally Forced Channel Flow

■ Initialization:

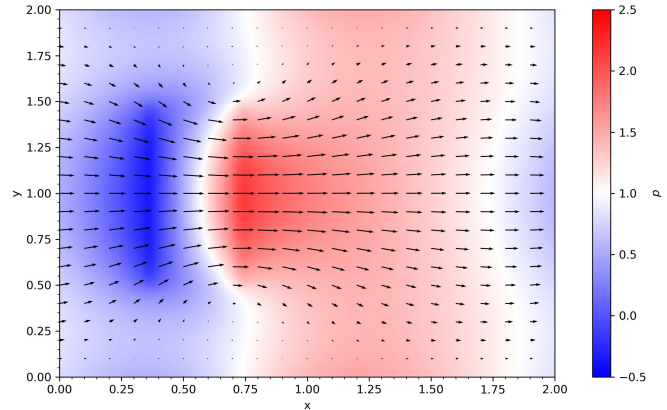
- vanishing velocity
- unity pressure
- rectangular source

■ Boundaries:

- Top & Bottom: $u = 0$ & $\nabla p = 0$
- Left & Right: periodic

■ Observation:

- central jet formation
- recirculation eddies



Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation
 - computationally efficient
 - highly accurate for simple problems
- Drawbacks:
 - struggles with complex geometries
 - strong numerical diffusion at low orders
 - inflexible rectilinear grid with fixed resolution
 - mass and energy can leak due to numerical errors

Integration Techniques

■ *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
- *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
 - ▶ used to model transport with defined direction
- *Central Difference:* considers both directions, more precise but can lead to oscillations
 - ▶ used to model symmetric dispersion processes
- *Multi Point Stencil:* higher order central method for computing ∇^2 in multiple dimensions

■ *Temporal:*

- *Backward Euler:* looks to future to calculate next step implicitly by inversion, extremely stable
- *Forward Euler:* looks to past to find next step explicitly, simple but unstable for large intervals
- *Leapfrog:* reaches back to previous step instead of current one, symplectic and invertible

Integration Techniques

■ Advanced:

- *Lax-Wendroff*: take intermediate step, equivalent to series expansion to second order
- *Runge-Kutta*: take trial steps to estimate curvature, then make informed jump
- *Predictor-Corrector*: take roughly predicted step, then recalculate with new information
- *Leapfrog*: preserve phase-space volume and total energy, enable backward reconstruction

■ *Courant-Friedrichs-Lewy Condition*: $\frac{v\Delta t}{\Delta x} \leq 1$

■ *Pure Diffusion Condition*: $\frac{2v\Delta t}{\Delta x^2} \leq 1$

Discretized Formulation

■ *Convection:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$

■ *Diffusion:* $\frac{\partial u}{\partial t} = v \frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} = v \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$

■ *Bateman-Burgers Equation:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = v \frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = v \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$

■ *Laplace Equation:* $\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \longrightarrow \frac{p_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{\Delta x^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{\Delta y^2} = 0$

Overview

Physical Background

- Fundamental Definitions
- Basic Equations
- Examples

Numerical Approaches

- Finite Difference Method
- Finite Volume Method
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Practical Implementation

- Kelvin-Helmholtz Instability
- Rayleigh-Taylor Instability
- Outlook

Conceptual Overview

- Idea: calculate flux through faces of control volumes that divide the domain using integral form

$$\frac{\partial}{\partial t} \int_V Q dV + \oint_S (F \cdot n) dS = \int_V \Sigma dV$$

- Advantages:
 - inherently strictly conservative for total mass and energy
 - unstructured grid handles complex geometries well
 - easily deals with shock discontinuities
- Drawbacks:
 - mathematically complex at higher order accuracy

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Conceptual Overview

- Idea: optimize polynomial shape function to approximate solution across whole grid elements
- Advantages:
 - highly mathematically robust
 - precise gradient calculation at boundaries
 - handles extremely complex mesh geometries well
- Drawbacks:
 - computationally expensive and memory intensive
 - bad stability for fast turbulent flows

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Setup

Visualization

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Setup

Visualization

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

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■ Frameworks:

- *Smoothed Particle Hydrodynamics*: mesh free interactions based on kernel weighting
- *Lattice Boltzmann Method*: cells simulate statistical particle collisions and streaming
- *Adaptive Mesh Refinement*: grid resolution increases where higher precision is required

■ Regimes:

- | | |
|--|---|
| ■ <i>Plasma</i> : use magnetohydrodynamics | ■ <i>Degeneracy</i> : need quantum effects |
| ■ <i>Relativity</i> : use relativistic hydrodynamics | ■ <i>Collisionless</i> : need initial distributions |

■ Applications:

- | | | |
|---------------------|-------------------|--------------------|
| ■ stellar evolution | ■ accretion disks | ■ cosmic filaments |
|---------------------|-------------------|--------------------|

Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*¹ 2022
- P. Mocz, *Medium*² 2020-2025

Resources on GitHub: [here](#)